

- ③ what are the basic elements of cache Diagram
- ④ what is the general relationship among access time, memory cost, and capacity?
- ⑤ A set associative cache contains of 64 lines, or slot, divided into four line sets. Main memory contains 4K blocks of 128 word each. show the format of main memory address.
- ⑥ Explain Direct mapping cache organization with diagram

[3]

1. Cache address
2. Size
3. Mapping function
4. Replacement algorithm
5. Write policy
6. Block size
7. Number of caches.

Processor	L-1	L-2	L-3
Pentium	8KB/8KB	256KB	*
Pentium Pro/600	32KB	256KB	3
Core i3	32KB	256KB	3 MB
Core i5	32KB	256KB	6 MB
Core i7	32KB	256KB	12 MB

Computer Organization and Architecture

Cache Memory

- ① # Compare direct, associative and set associative memory. Explain the replacement algorithm of cache memory.
- ② Suppose FFFFF8 is an address of 16M memory. Find out the corresponding address of 16K line cache in associative mapping.

$$MM = 24 \text{ bit} = 16 \text{ MByte}$$

$$CM = 64 \text{ KB} = 16 \text{ bit}$$

MM 24 bit where 2 bit (LSB) for identifying [2 bit word] individual byte with the cache block. [22 bit block identifier] 22 bit (MSB) for Tag. [8 bit tag and 14 bit slot or line]

Main Memory add 24 bit

F F F F F 8
1111 1111 1111 1111 1111 1000

For associative mapping (tag 22 bit) and (word 2 bit)

∴ Tag (s) 22 bit → 1111 1111 1111 1111 1111 10

Word (w) 2 bit → 00

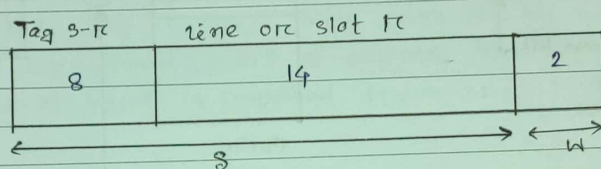
In cache memory line is not considerable and any block of main memory can be mapped in any line of CM.

Tag 0011 1111 1111 1111 1111 1110
cache add → - 3 F F F F E
Word 00

[6]

Direct Mapping:

1. Each block of main memory maps to only one cache line.
→ If a block is in cache, it must be in one specific place
2. Address is in two parts
3. Least significant w bit identify unique word.
4. Most significant s bit specify one memory block.
5. Cache line $(s-w)$ Tag
- 6.



6. No two blocks in same line have the same tag field

Direct Mapping Pros and Cons

Pros: 1. Simple

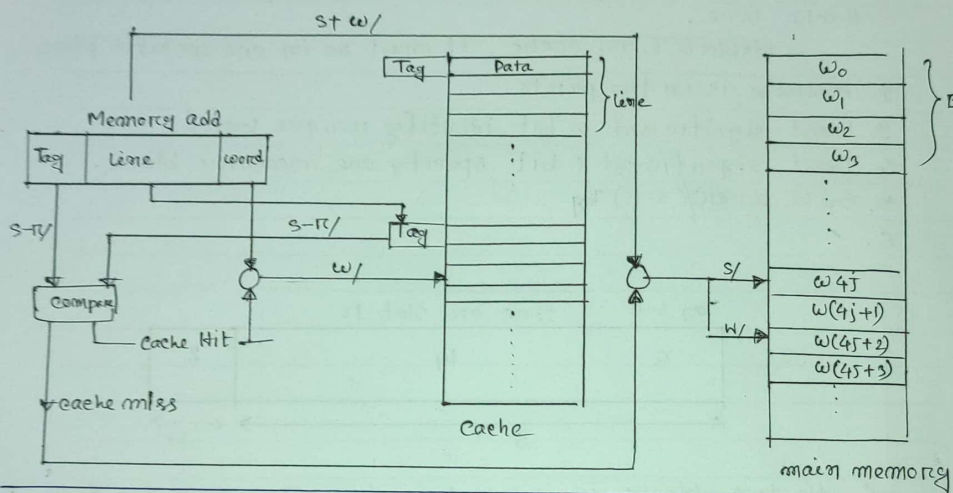
2. Inexpensive

Cons: 1. Fixed location for given block

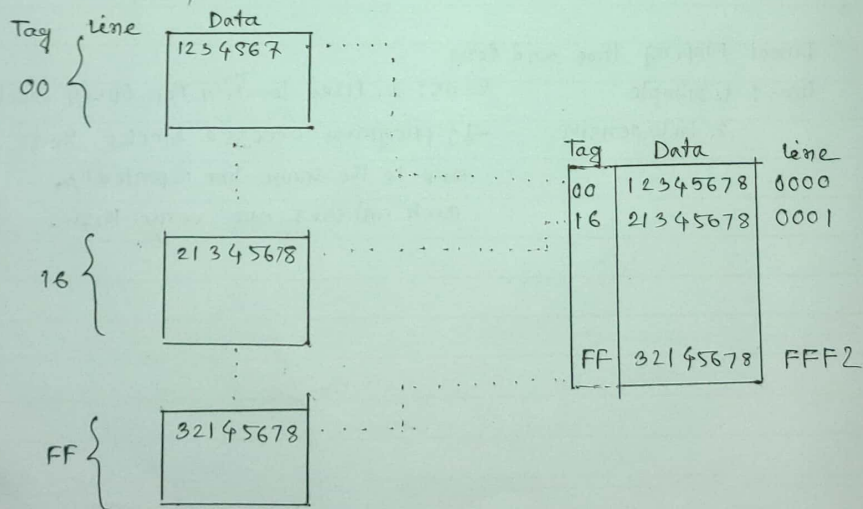
- If program access 2 blocks that map to the same line repeatedly.

each misses are very high.

W Direct cache mapping organization.



W Direct Mapping Example:



16 MByte memory

Associative Mapping:

1. A main memory block can load into any line of cache.
2. Memory address is interpreted as tag and word.
3. Tag uniquely identifies block of memory.
4. Every line tag is examined for a match.
5. Cache searching gets expensive.

Tag 22 bit	word 2
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6. 22 bit tag stored with each 32 bit block of data
7. lead significant 2 bit of address identify which 16 bit of word is required from 32 bit data block

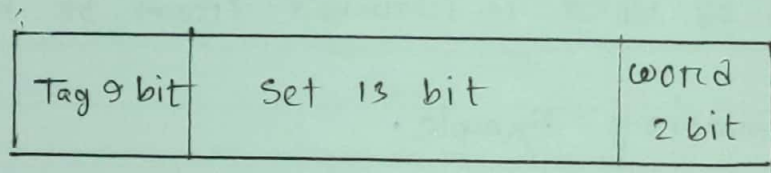
Associative mapping Example.

Add.	Data
1 2 3 4 5 6	1 2 3 4 5 6 7 8
2 3 4 5 6 7	2 1 3 4 5 6 7 8
	32 bit

Tag	Data	line No.
0480 15	12345678	0000
	21345678	0001
22 bit	32 bit	

□ Set Associative Mapping

1. cache is divided into a number of sets
2. Each set contains a number of lines
3. a given block maps to any line in a given set.
4. 2 line per set
- 2 way associative mapping



⑤

No of lines = 64

These are divided into 4 line sets.

so, number of sets = $64/4 = 16$

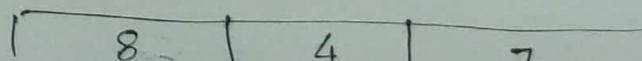
$$\text{bit per set} = 2^4 = 16 \text{ (4 bit)}$$

Number of 4k block = 2^{12} blocks (12 bit block no)

$$\therefore \text{Tag} = (\text{block (bit)} - \text{set})$$

$$= 12 - 4 = 8$$

No of bit for word = $2^7 = 128 = (7 \text{ bi})$



Two way Set associated Mapping

Tag	Set No	Data
000	0000	12345678

	Tag	Data	Set No	Tag	Data
015	0000	00000000		000	12345678
		23456789		015	23456789
		11111111		018	12341234
					11111111

9 bit 32 bit

9 bit 32 bit

018	0000	
		12341234

32 bit

□ Replacement algorithm: (For high speed acquire)

- For direct mapping:
1. No (replacement) choice
 2. Each block only map one time
 3. replace that line

Replacement algorithm (Asso / set Asso)

1. Hardware implemented algorithm
2. least Recently used
3. First in First out
4. least Frequently used
5. Random